① Block Diagram

Input

↓

Coupling Capacitor 1

↓

gain 10x — CE stage → Bypass Capacitor 1 → Coupling Cap 2

↓

gain 1x — CC buffer

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Coupling Cap 3

↓

gain 10x — CE stage → Bypass Capacitor 2 → Coupling Cap 4

↓

gain 1x — CC buffer

↓

Coupling Capacitor 5

↓

Load → output

total gain = $10 \times 1 \times 10 \times 1 = \boxed{100}$

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Calculations ;

for $1^{st}$ CE stage ;

$$\text{gain} = \frac{V_o}{V_i} = \frac{x}{10m} = 10 \quad \leftarrow x = 10 \times 0.01$$

$$x = 0.1\,mV$$

* Small signal resistance needed

$$r_e = \frac{26\,mV}{I_E} = \frac{26\,mV}{1\,mA} = \boxed{25 \text{ ohm} - 26}$$

$$\text{gain} = \frac{-R_C}{r_e + R_E}$$

let $R_E = 390$

then $R_C = \boxed{4150 \text{ ohm} \simeq 4200}$

* $V_E = I_E R_E \Rightarrow V_E = 1mA \times 390 = 0.39$

$V_{BE} = 0.7$ * BJT

then $V_B = 0.39 + 0.7 = 1.09V$.

* $V_C = V_{CC} - I_C R_C$ → let $\beta = 100$

* $I_C \simeq I_E$ then $I_B = \frac{1mA}{100} = 10\mu$

$V_C = 12V - 1mA \times 4200 = 7.8V$

* $V_{CE} = V_C - V_E = 7.8 - 0.39 = 7.41V$

* Divider current ; $10 I_B = I_{div}$

$\Rightarrow I_{div} = 100\mu A$

Total div R ;

$$R_1 + R_2 = \frac{12}{100\mu} = 120k \text{ ohm}$$

$$* V_B = \frac{V_{CC} \times R_2}{R_1 + R_2} = \frac{12 \times R_2}{120k} = 1.09 \quad \leftarrow \boxed{R_2 = 10.9k \simeq 10k, \; R_1 = 110k}$$

For maximum swing : $V_{CE} = \frac{V_{CC}}{2} = 6V$

ours is $7.41 > 6$ (safe)

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for 1st CC stage ;

* $V_B \simeq 6V$

$V_E = V_B - 0.7$

$= 6 - 0.7 = 5.3V$

* $\boxed{\text{let } R_E = 1k}$

* $I_E = \frac{V_E}{R_E} = \frac{5.3}{1k} = 5.3mA$

* $I_E \simeq I_C$

so $I_C \simeq 5.3mA$

* $I_B = \frac{I_C}{\beta} = \frac{5.3mA}{100} = 53\mu A$

* $I_{div} = 10\, I_B$

$= 530\mu A$

$R_1 + R_2 = \frac{V_{CC}}{I_{div}} = \frac{12}{530\mu} = 22.6k\text{ohm}$

* $V_B = \frac{V_{CC} \times R_2}{R_1 + R_2} \Rightarrow 6 = \frac{12 R_2}{22.6k}$ so $\boxed{R_2 = R_1 = 11.3k}$

$\simeq 11k\text{ohm}$

* $A_V$ should be 1

$A_V = \frac{R_E}{R_E + r_e} = \frac{1000}{1000 + 4.7} = 0.995 \simeq 1$

$r_e = \frac{25mV}{I_E} = \frac{25mV}{5.3m} = 4.7\text{ ohm}$

* $Z_{out} = \frac{R_E}{\beta} = \frac{1000}{100} \simeq 10\text{ ohm}$

* $Z_{in} = (\beta + 1) R_E = (101)\, 1k = 101k\text{ ohm}$

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